

JKBOSE

🔥 2026 CLASS 11th 🔥

**UNIT AND
MEASUREMENTS**

PHYSICS LECTURE-1

BY-WAJAHID SIR



TOPICS TO BE COVERED

- 1 Physical Quantities
- 2 Basic Physical Quantities
- 3 Derived Physical Quantities
- 4 Supplementary Physical Quantities

Tumhe Kis Cheeze ka ddhyan dena hai

- Have Patience ✓
- Attain all the class & follow it blindly ✓
- Koi class nahi chorni ✓
- Last takk tike Rehna ✓
- Notes → 2 Pen/Pencil ✓



Main Kya krne wala hu

- Basic → Advanced
- Will give you Best *Best test*
- Explanation + Notes (Relevant Content)
- Book Question + Exercise + PYQ
- 8th & 9th Math → Jitna physics main zaruri hai ✓ *(Numericals)*
_{10th}



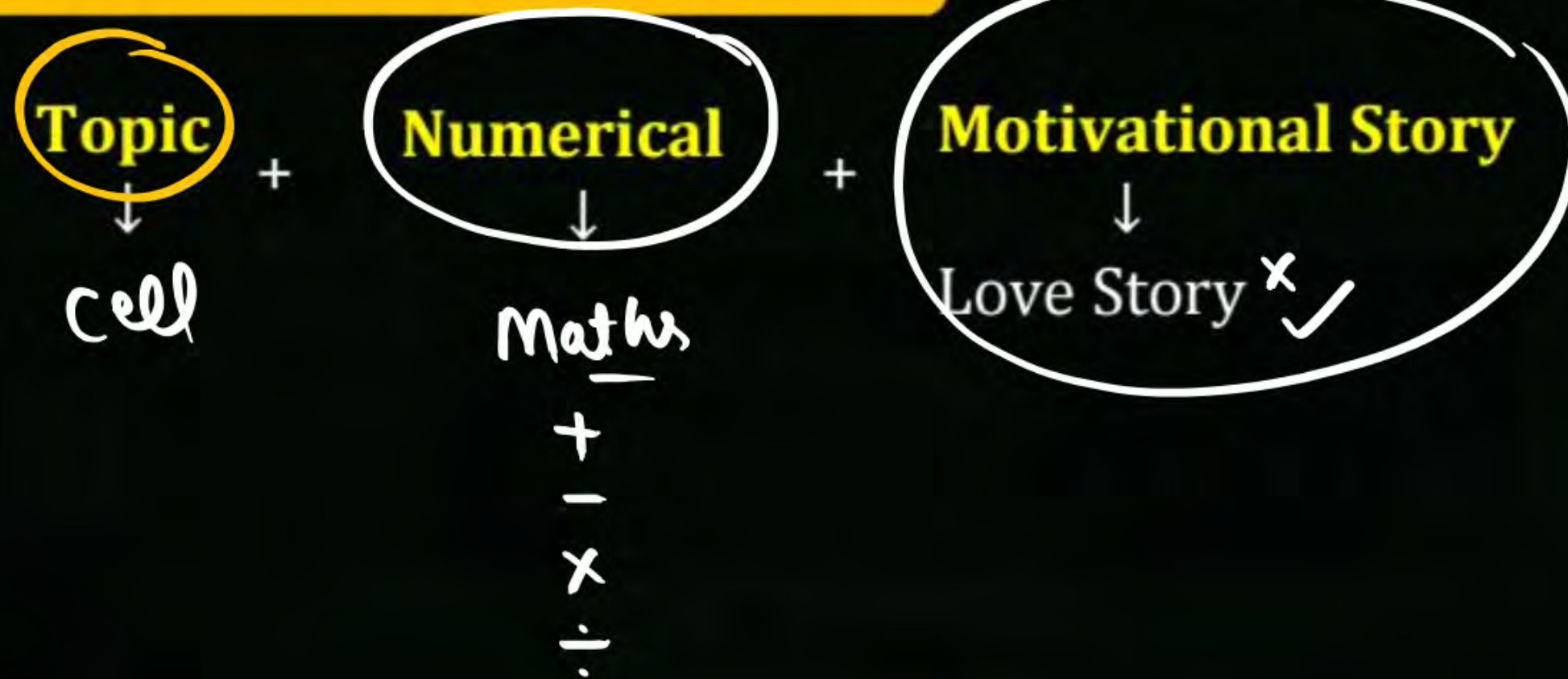
Problem

- Concept understanding (Easy Language)
- Fear of Physics
- Calculation → Pura Maths ✓
- Numericals ka dar ✓

Hereo
Ladale ♥



Padhne Ka Style



1. Class → Listen

2. Practice

3. Time Table * *



Syllabus

1. Unit & Measurement ✓
2. Motion in a straight line ✓
3. Motion in a plane ✓
4. Law of motion ✓
5. Work energy and power ✓
6. System of P & R ✓
7. Gravitation ✓
8. Mechanical properties of solids ✓
9. Mechanical properties of fluids ✓
10. Thermal Properties of Matter ✓
11. Thermodynamic ✓
12. Kinetic theory ✓
13. Oscillation ✓
14. Waves ✓



Unit I: Physical World and Measurement

05 marks

Units and Measurements

Need for measurement: Units of measurement; systems of units; SI units,
fundamental and derived units. Significant figures. Dimensions of physical quantities,
dimensional analysis and its applications. ✓

Introduction of Science

5th
↓
EVS
6th
↓
Science

Science:- is the systematic Knowledge gain through

Experiment & observation

gravity

9th

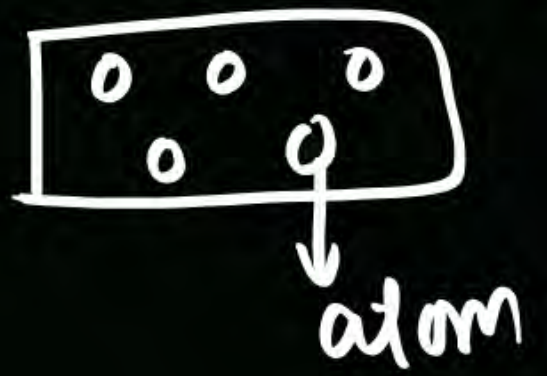
(Physics)
Physics

Vishal Sir
Chemistry

Dawood Sir
Biology

Basic
• Study of Pa
measure
Mass = 70kg
Length = 2m

• Study of Matter



• Study of Jaan (life)

Plants
↓
Botany

Animals
↓
Zoology

Science

↓
Experiment
+ observation

↓
Knowledge

mind

Q



?



Spherical
Shape?



Physical Quantities

PQ

→ Study → Physics

pq

The quantity you can measure is called

Example:-

Mass = 70kg Energy = 50J Length = 5m

Time = 1.5hr

Temperature = 23°C

Non-Physical Quantity:- The Quantity which you cannot Measure. ✓

Love = 50kg X

Emotion X

Hate X



$PQ = \text{Measure}$

✓ $\text{Mass} = \underline{40\text{kg}}$

$\downarrow \downarrow \downarrow$
 $PQ = nu$

2092
P4Q

✓ Representation
of PQ .

$\text{Length} = 100\text{m}$

$PQ = nu$

$\text{Time} = 60\text{sec}$

$PQ = nu$

$PQ = nu$

$\downarrow \uparrow \frac{PQ}{u} = n \uparrow \downarrow$

Inversely Related

Important.

Physical Quantities

$$PQ = nu$$

Physical Quantity

Experiment & Observation

① Basic/Fundamental PQ

② Derived PQ

③ Supplementary PQ

①

Basic Physical Quantities

- Fundamental PQ.
- Physics starts from here
- 7 Basic PQ.

*
Zada
dayaar
dena
hai

- ① Mass. $\rightarrow$ kg $\rightarrow$ weighing machine
- ② Length. $\rightarrow$ metre $\rightarrow$ Scale
- ③ Time $\rightarrow$ Second $\rightarrow$ Watch.
- ④ Current $\rightarrow$ Ampere $\rightarrow$ Ammetre
- ⑤ Temperature $\rightarrow$ K $\rightarrow$ Thermometre.
- ⑥ no. of substance $\rightarrow$ mole $\rightarrow$ H/W
- ⑦ luminous intensity $\rightarrow$ Candella $\rightarrow$ H/W.



② Derived Physical Quantity

DPO

The PQ which is derived from Basic/Fundamental Physical Quantity.

① Area = L × B = m × m = m²

② Volume = L × B × H = m × m × m = m³

③ Density = $\frac{m}{V}$ = $\frac{kg}{m^3}$

④ Speed = $\frac{\text{distance}}{\text{time}}$ = $\frac{m}{s}$

⑤ Velocity = $\frac{\text{displacement}}{\text{time}}$ = $\frac{m}{s}$



$$\textcircled{6} a = \frac{\textcircled{v}}{\textcircled{t}} = \frac{m}{s \times s} = \textcircled{\frac{m}{s^2}}$$

$$\textcircled{7} F = \textcircled{m} \textcircled{a} = \textcircled{kg} \textcircled{\frac{m}{s^2}} = \underline{\text{Newton}}$$

Note

Kabhi Kabhi

unit Scientist K
naam pa b

Rakhi jati hai

$$\textcircled{8} \text{ momentum } (p) = mv \\ = kg \cdot m/s$$

$$\textcircled{9} \text{ Work} = f \cdot d \\ = \textcircled{kg \frac{m}{s^2}} \times m \\ = \textcircled{kg \frac{m^2}{s^2}} = \underline{\text{Joule}}$$

$$\textcircled{10} \checkmark P = \frac{F}{A} = \frac{N}{m^2} = \text{Pascal}$$

$$\textcircled{11} \text{ Power} = \frac{1}{f} = \frac{1}{m'} = m' = \text{Dioptre}$$

Tough.

3

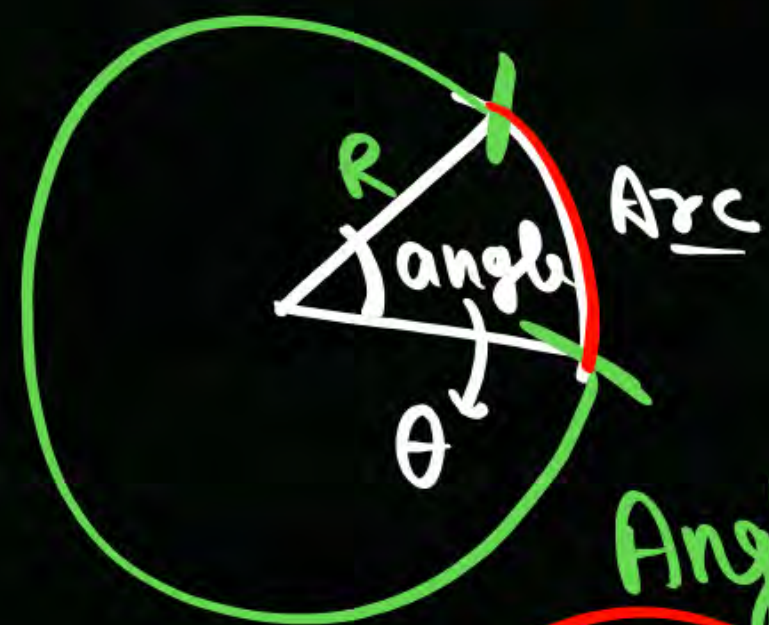
Supplementary Physical Quantities

- This Physical Quantity do not have ~~units~~ dimensions

Lecture 2

Zabardasti → Units

1. Angle



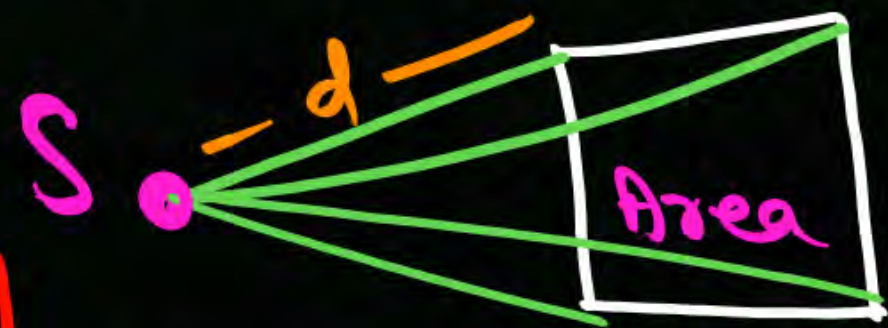
Basic Math

$$\text{Angle} = \frac{\text{Arc}}{R} = \frac{m}{m} = 1$$

degree | Radian

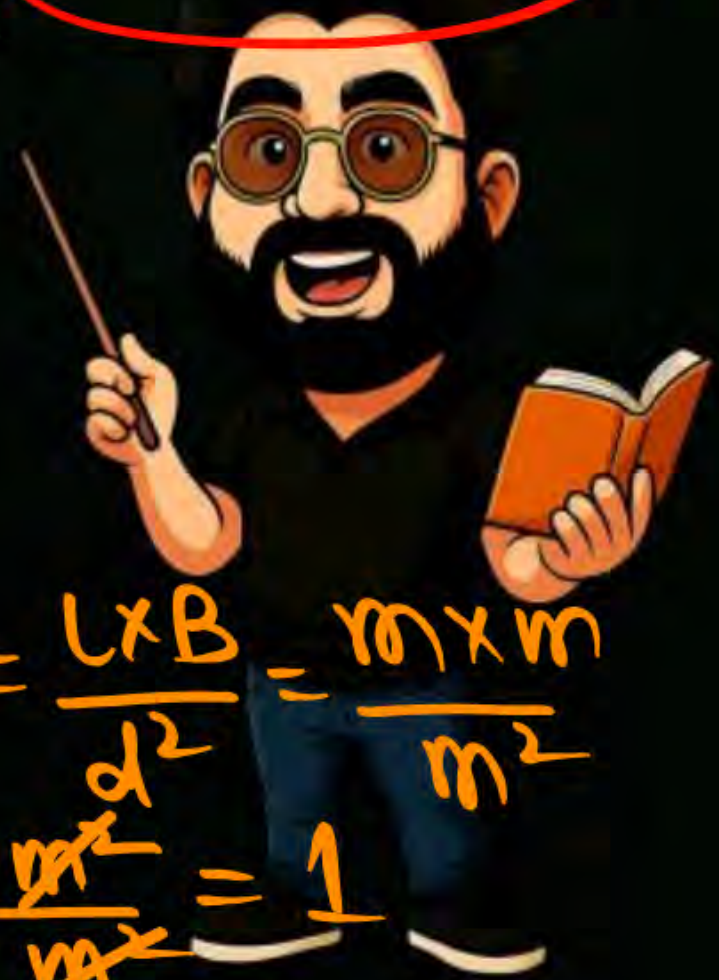
2. Solid Angle

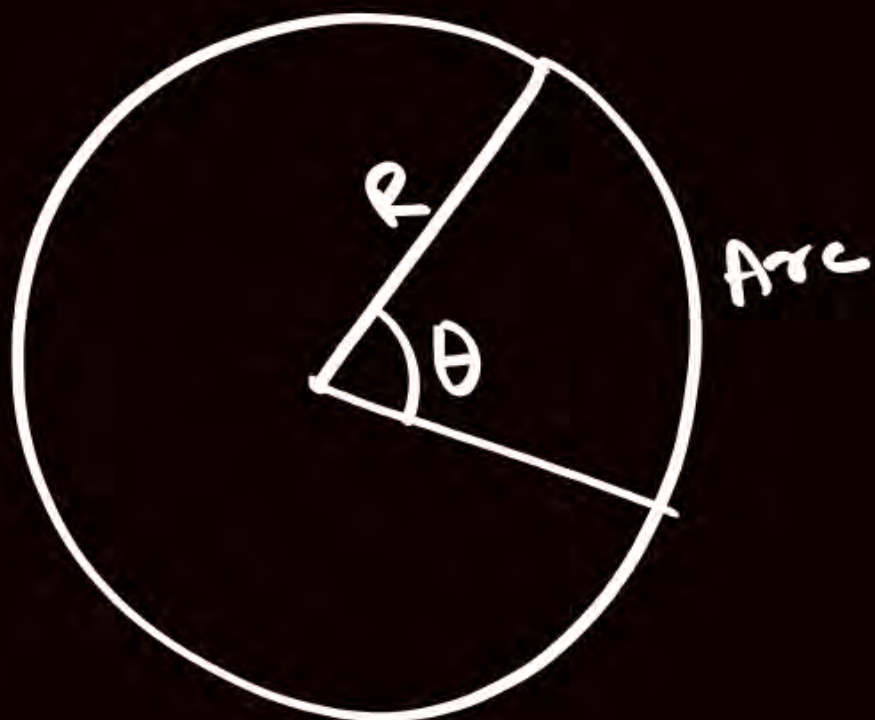
- the angle made when you see the Area



$$\begin{aligned} \text{Solid Angle} &= \frac{\text{Area}}{d^2} = \frac{L \times B}{d^2} = \frac{m \times m}{m^2} \\ &= \frac{m^2}{m^2} = 1 \end{aligned}$$

Steradian

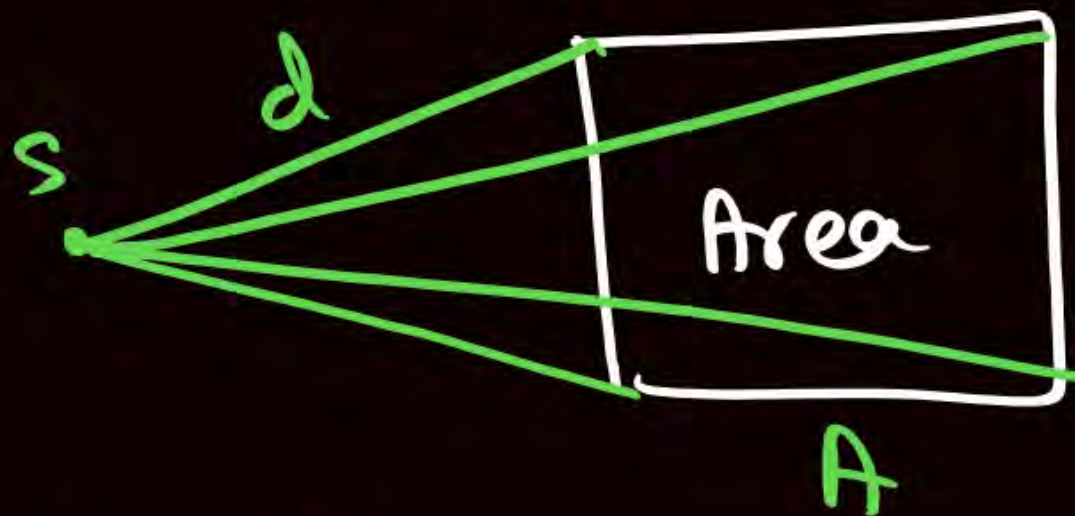




$$\theta = \frac{\text{Arc}}{r}$$

$$\theta = \frac{\pi}{\pi} = 1$$

° | Radian



$$\begin{aligned} SA &= \frac{A}{d^2} \\ &= \frac{L \times B}{m^2} = \frac{m \times m}{m^2} = \frac{m^2}{m^2} = 1 \end{aligned}$$

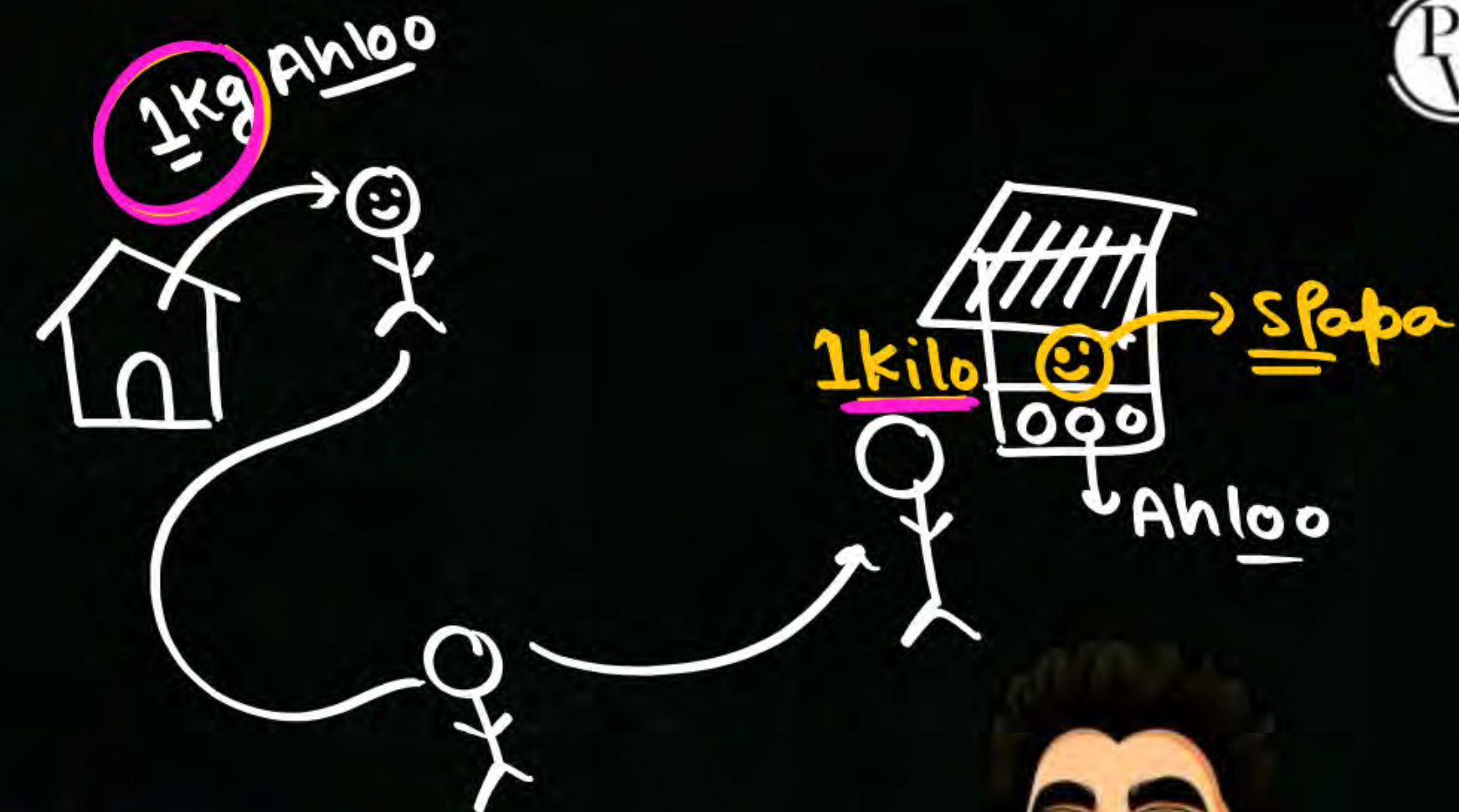
Steradian

Number name

the name of the number

$$\underline{\underline{1\text{Daron}}} = \underline{\underline{12}}$$

$$\underline{\underline{1\text{kilo}}} = \underline{\underline{1000}}$$



$$\underline{\underline{1\text{kilo}}} = \underline{\underline{1000}}$$



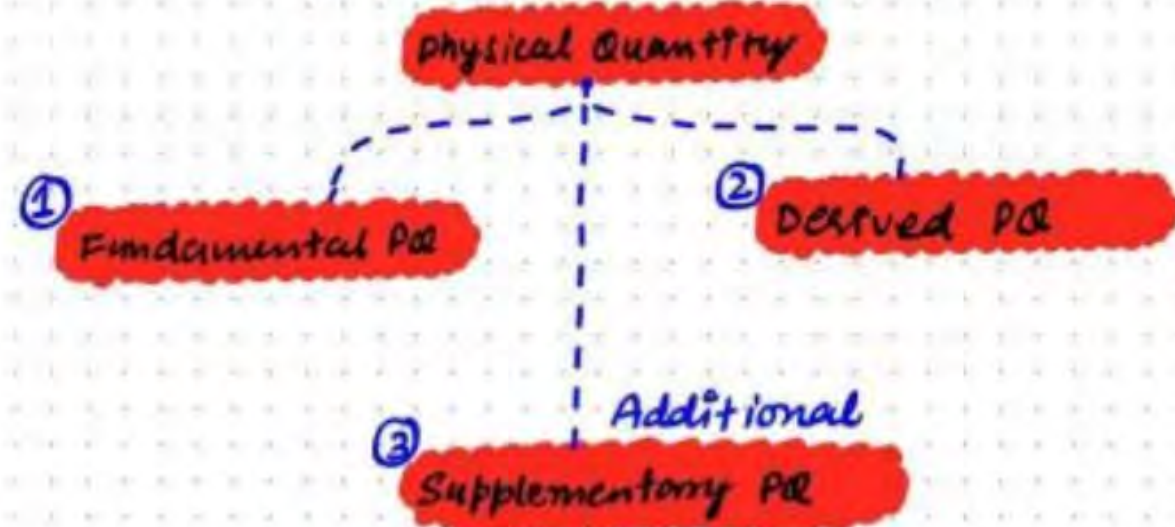


Physics:- the Branch of Science which deal with the study of the physical Quantity

physical Quantity the Quantity you can measure is called physical Quantity

Mass = 4kg Length = 2m Time = 2sec

$$PQ = nu$$



DEKH Bhai Physics Numerical k Bina khuch be nahi hai

Physics KA matlab **NUMERICAL**



Fundamental Physical Q

the Basic Physical Quantity which donot depend upon the other Physical Quantity

There are Seven(7) Physical Quantity

PQ	unit	old unit
Mass	kg	M
length	m	L
time	s	T
temp	Kelvin(K)	K
current	Ampere(A)	A
no. of Substance	mole	mole
Luminous Intensity	candella(cd)	cd

Derived Physical Q

the Physical Quantity derived from one Basic Physical Quantity

$$\text{Speed} = \frac{\text{distance}}{\text{time}} = \text{m/s}$$

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}} = \text{m/s}$$

$$\text{acceleration} = \frac{\text{Velocity}}{\text{time}} = \frac{\text{m/s}}{\text{s}} = \text{m/s}^2$$

$$\text{Force} = m \times a = \text{kg m/s}^2$$

Samajh Main Aya Teko



Supplementary Physical Q

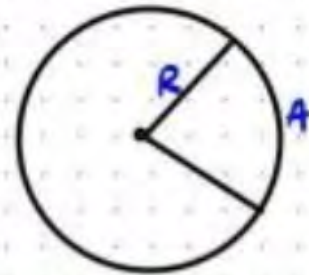
Lecture no. 2

②

There are two additional Physical Quantity

Plane Angle θ

It is the Ratio of Arc to the Radius



$$\theta = \frac{\text{Arc}}{\text{Radius}} = \frac{m}{m} = 1$$

unit = Radian
unit = degree

$$\pi \text{ rad} = 180^\circ$$

Solid Angle

It is the Ratio of Area to the Square of distance from point of observation



$$\text{Solid Angle} = \frac{\text{Area}}{R^2} = \frac{m^2}{m^2} = 1$$

Steradian

Bool
Dubara
Samjao





Question

Derive an expression for velocity and acceleration of a particle executing simple harmonic motion.

DISPLACEMENT/VELOCITY/ACCELERATION/FORCE

Let us consider SHM (Oscillation).

Displacement

$\omega = \frac{\theta}{t}$
 $\omega t = \theta$

$x = A \sin \omega t$
 $-A \cos(\omega t + \phi)$

Diagram illustrating SHM using a circular reference:

Labels in diagram: ω , EP , MP , A , ϕ , $Phase$, $A = r$.



Question

22. Discuss the harmonics in the case of open end organ pipes.

Standing wave ✓

fixed at two ends
in organ Pipe





Question

Explain modes of transfer of heat with example.

MODES OF TRANSFER OF HEAT

Conduction

Convection

Radiation



Question

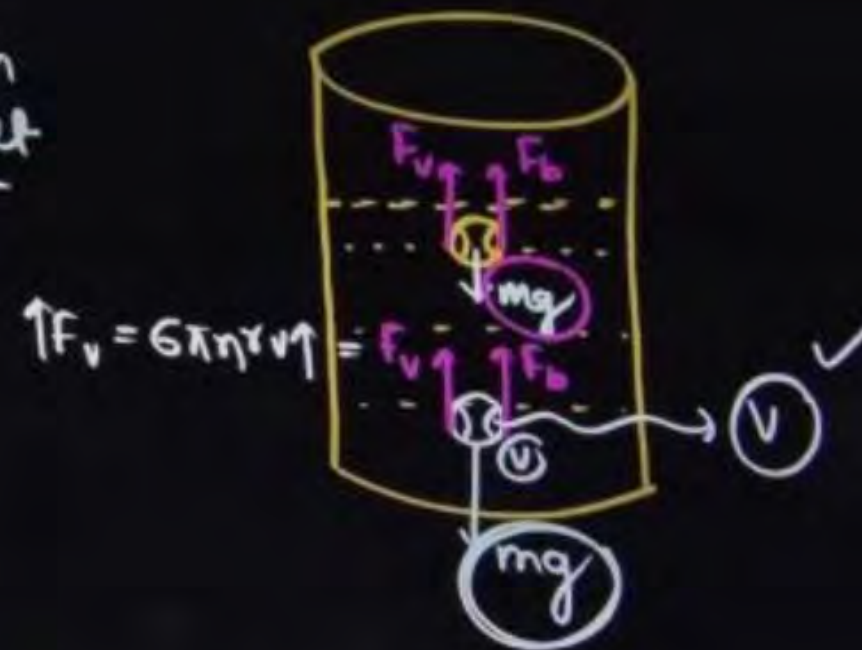
21. What do you mean by terminal velocity? Derive an expression for it.

Terminal Velocity

maximum const velocity An
on reach when object
through liquid

$$\underline{W} = F_v + F_b$$

$$mg = 6\pi\eta r v + mg$$



18:50



Question

From the velocity-time graph of uniform accelerated motion, deduce the equations of motion in

- (i) Velocity and time (iii) Position and time

$$(1) \underline{v = u + at}$$

$$\tan \theta = \frac{P}{B}$$

$$= \frac{y}{t}$$

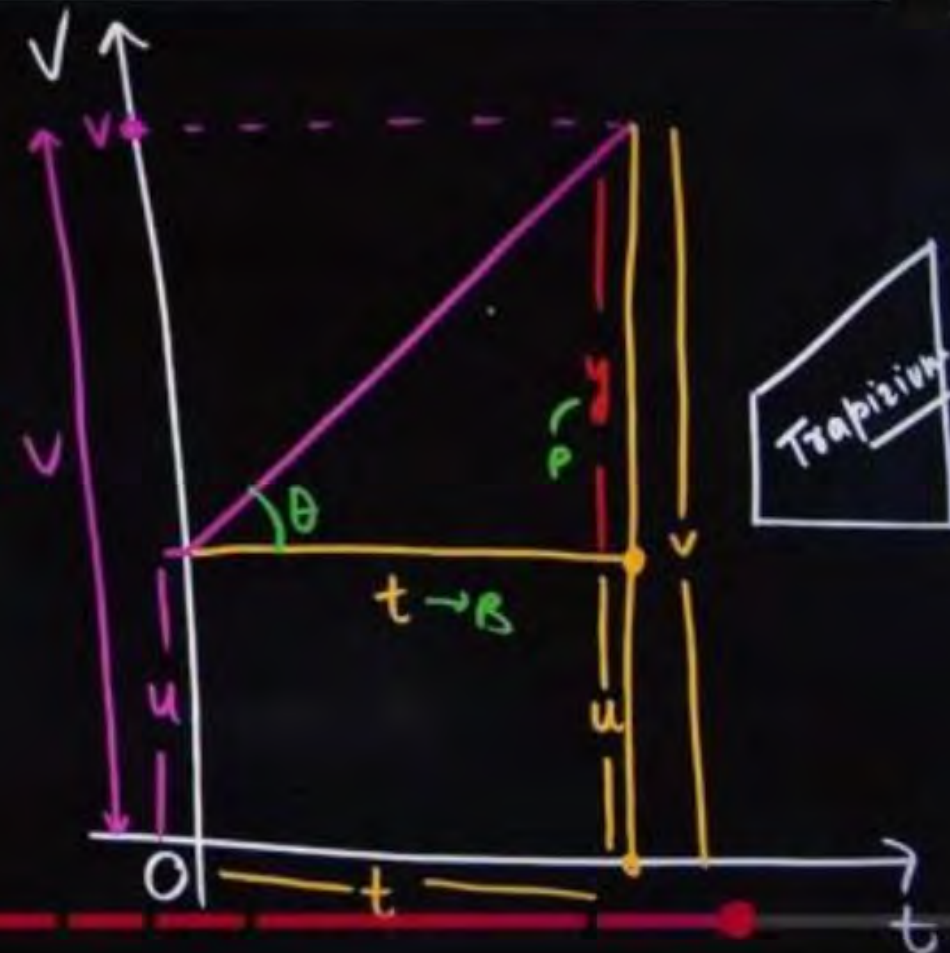
$$a = \frac{v-u}{t}$$

$$a = \frac{v}{t}$$

$$v = y + u$$

$$v - u = y$$

$$y = v - u$$





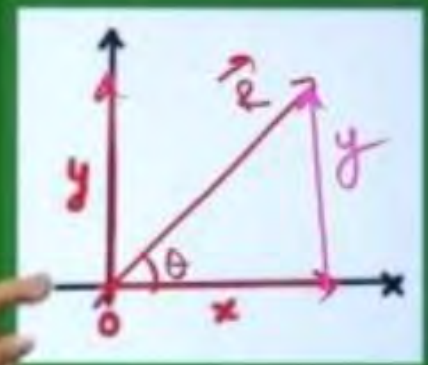
Question

20. What do you mean by resolution of a vector? Find rectangular components of a vector.

Component of Vector

► Converting the Resultant vector into

• $\vec{r} = \frac{r}{\sqrt{2}}(\hat{i} + \hat{j})$



MOSEVILLAH



Question

19. At what distance from the mean position is kinetic energy equal to potential energy? Briefly explain in case of S.H.M.

19. At what distance from the mean position is kinetic energy equal to potential energy? Briefly explain in case of S.H.M.

19:25

$$\frac{A}{\sqrt{2}}$$

2025

Sol K.E =

$$K.E = \frac{1}{2} m A^2 \omega^2 x^2$$

$$\frac{1}{2} k x^2 + \frac{1}{2} k x^2$$

$$x$$



Question

18. Find an expression for the gravitational potential energy of a body situated at a distance " r " from the centre of earth.

GRAVITATIONAL POTENTIAL ENERGY

- ✓ Work done in moving a particle from infinity to a distance r from the centre of the earth is the gravitational potential energy of the particle.
- ★ The mass of the particle is m and the unit of potential energy is J kg^{-1} .





Question

16. Distinguish between elastic and inelastic collisions.

Collision

13:03

When two or more bodies interact and exert the force on each other

Elastic Collision

✓ K.E

✓ P

Inelastic

Body Stick



Question

14. State Newton's first and second laws of motion, also define one Newton of force.



Newton's 2nd Law of Motion

• Magnitude of Force

• Force Formula ✓

The rate of change of momentum

$$F = \frac{\Delta P}{\Delta t}$$

$$F = \frac{dP}{dt}$$

$$F = \frac{dmv}{dt}$$

Δ = bad

d = chotu ch

$F = ma$



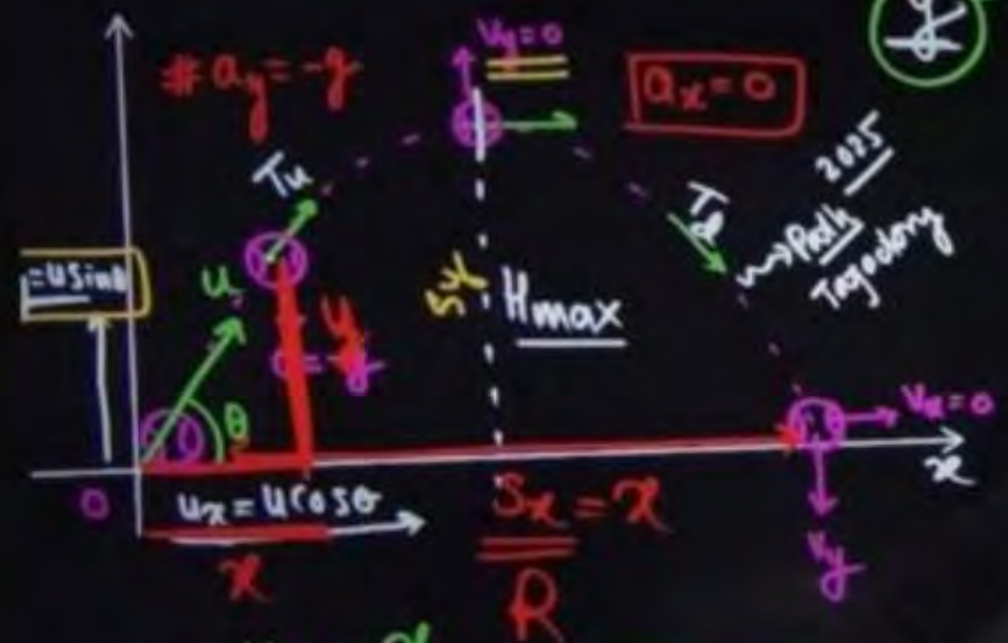


Question

12. Derive an equation of trajectory of a projectile fired at an angle.

(4) Equation of Trajectory

the Relationship b/w $\frac{x}{y}$



$$\# u_x = \frac{x}{t}$$

$$t = \frac{x}{u_x}$$

$$t = \frac{x}{u \cos \theta}$$

$$S = ut + \frac{1}{2} at^2$$

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$y = u \sin \theta t + \frac{1}{2} (-g) t^2$$

$$y = u \sin \theta t - \frac{1}{2} g t^2$$

$$y = u \sin \theta t$$

$$= g \left(\frac{x}{u \cos \theta} \right)^2$$



Question

11. Write the uses and limitations of the dimensional analysis.

Limitation of Dimension

The limitations of dimensional analysis are:

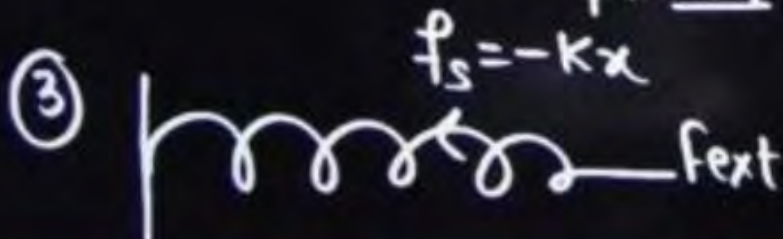
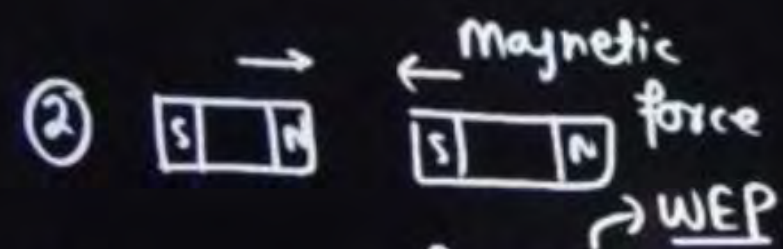
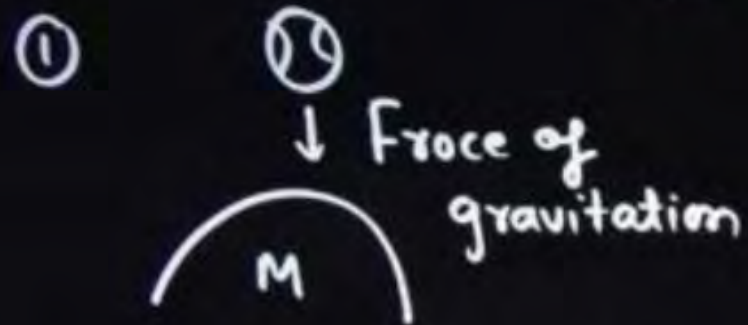
- 1. We cannot derive the formulae involving trigonometric functions, exponential functions, log functions etc., which have no dimension.
- 2. It does not give us any information about the numerical constants in a formula.
- 3. Dimension of a quantity is not unique.



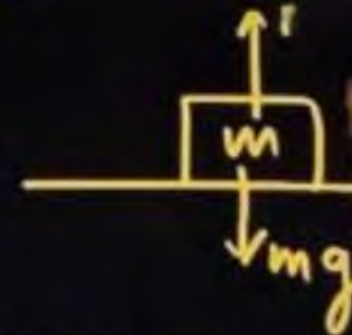
Question

10. What is meant by restoring force and force constant?

Types of forces



④ Contact force



⑤ Friction

Full screen



Question

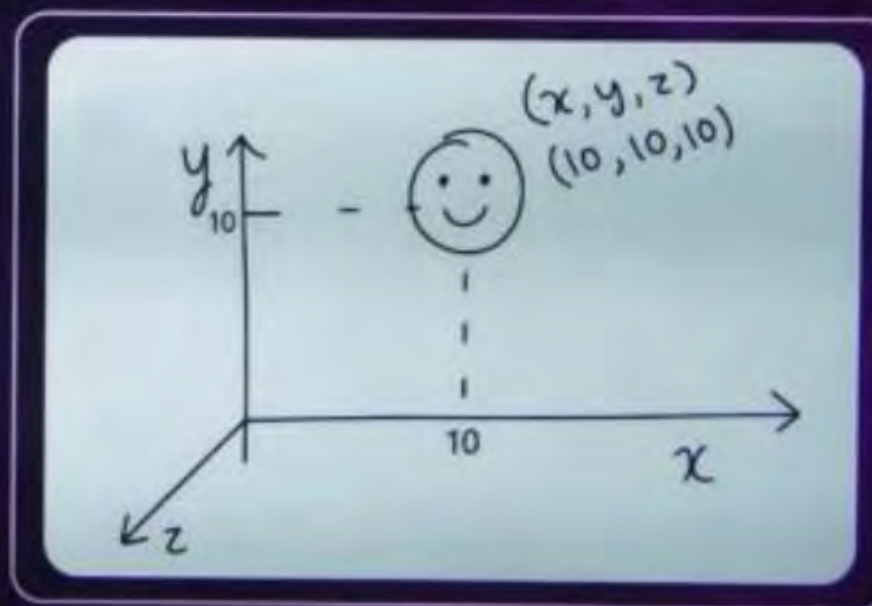
9. What is meant by degree of freedom? Write down the degree of freedom of diatomic gas molecule.

DEGREE OF FREEDOM

★ The total no. of co-ordinate required to specific the position of a molecule.

Motion of a molecule

De... om
 • He - ③
 • O₂ - ⑤
 ic gas → O₃ - ⑥





Question

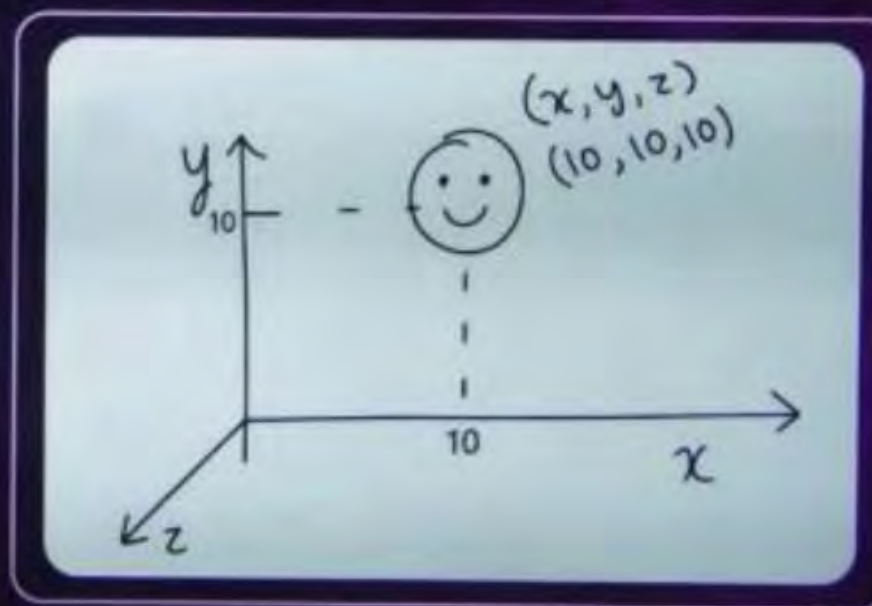
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DEGREE OF FREEDOM

★ The total no. of co-ordinate required to specific the position of a molecule.

Motion of a molecule

De... om
 • He - ③
 • O₂ - ⑤
 ic gas → O₃ - ⑥





Question

8. Write down the four assumptions of Kinetic theory of gases.

KINETIC THEORY OF GAS (KTG)

- ★ Gas is made up of free gas molecule.
- ★ They have velocity (K.E → kinetic energy, related temperature).

$$K.E = \frac{1}{2}mv^2$$
- ★ They are all in random motion.
- ★ Collision (elastic collision)
 - Each other
 - Wall
- ★ Collision (elastic collision)



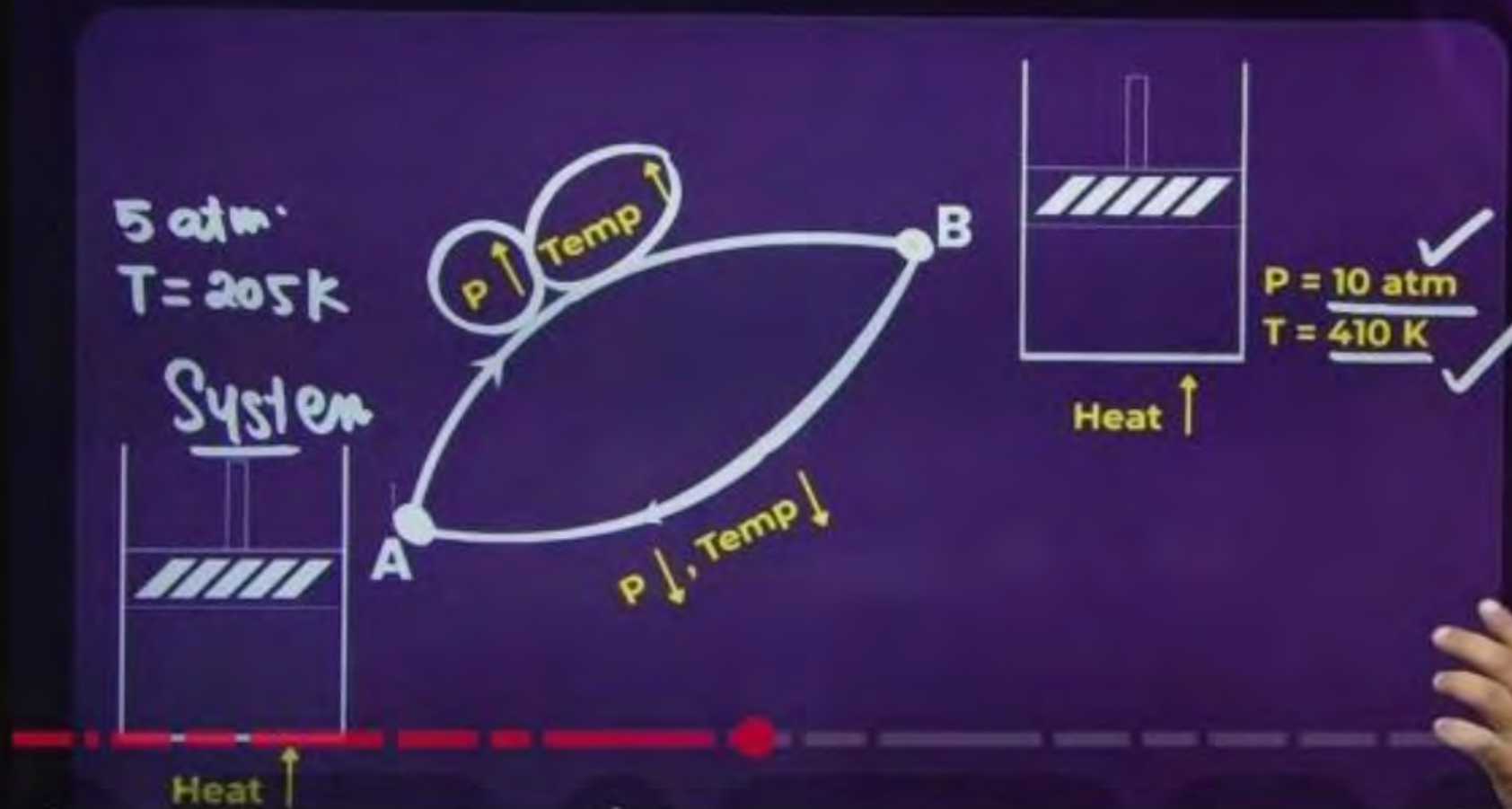


Question

Q. What are cyclic and non-cyclic processes?

CYCLIC PROCESS

(PYQ)





Question

5. Obtain the dimensions and units of coefficient of viscosity.

→ coefficient of viscosity

$$\frac{6\pi\eta r v}{r}$$

Ignore

$$\tau = \left(6 \times \frac{22}{7}\right) \eta$$

$$\frac{MT^{-1}}{L} = \eta$$

$$\boxed{ML^{-1}T^{-1} = \eta}$$

$$= \eta L^1 L^1 T^{-1}$$

$$MLT^{-2} = \eta L^2 T^{-1}$$

$$\frac{MLT^{-2}}{L^2 T^{-1}} = \eta$$

Physics With



Question

Q4. State the differences between Mass and Moment of inertia. (any two)

MOMENT OF INERTIA

★ The moment of inertia is a measure of resistant an object is to changes in motion.





Question

3. Write the equation for momentum, and work done in rotational motion.

KINEMATIC EQUATION OF ROTATIONAL MOTION



$$v = u + at$$

$$\omega = \omega_0 + \alpha t$$

$$s = ut + \frac{1}{2}at^2$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$



Question

2. State and establish the conservation law of angular momentum.

UP BOARD CLASS 11th PHYSICS | ROTATIONAL MOTION

www.youtube.com - To exit full screen, press Esc

LAW OF CONSERVATION OF MOMENTUM

↓ Const

When external torque is zero then initial angular momentum is equal to final angular momentum.

$\vec{\tau} = \frac{d\vec{L}}{dt}$

$0 \times \Delta t = \Delta \vec{L}$

$0 = \Delta L$

$0 = L_f - L_i$

$L_i = L_f$

L

Physics Wallah



Question

1. (i) The number of significant figure in 6.032 Nm^{-2} is:

(a) 6

(b) 3

(c) 2

(d) 4

Turn Over

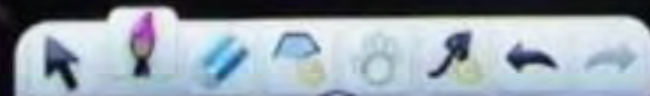
3. $\frac{2400}{\text{SF} = 2}$

4. 2

⑥ Exact Number
Counted Number
400 students.

⑦ $2.64 \times 10^{27} \text{ kg}$

PhysiWalla





Question

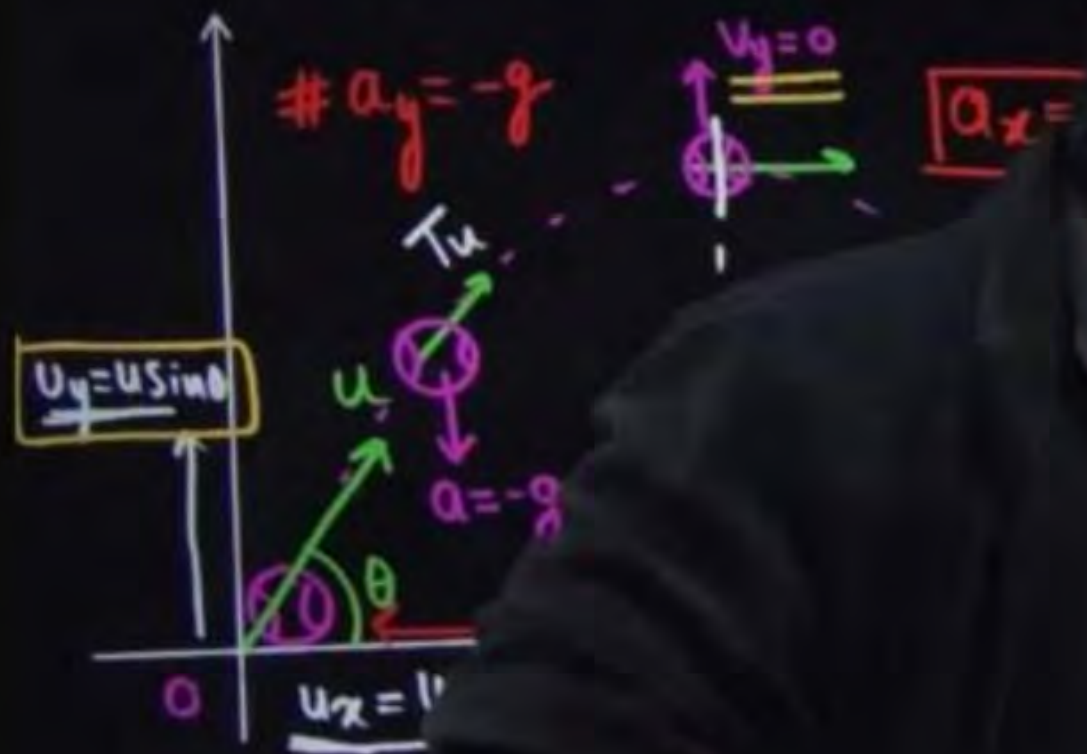
(iii) The horizontal range covered by the projectile is proportional to:

- (a) Its velocity
- (b) Square of the velocity
- (c) Sine of angle of projection
- (d) Square of sine of angle of projection

JKBOSE Class 11th Physics | Motion in a Plane | ONE-SHOT # Gun ...

② Range (R)

The Maximum Horizontal displacement



$$s = ut + \frac{1}{2}at^2$$

$$x = u_x t + \frac{1}{2}a_x t^2$$

$$x = u_x t + 0$$

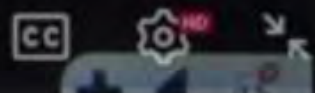
$$R = u_x \times \text{Time of flight}$$

$$R = u \cos \theta \times \frac{2u \sin \theta}{g}$$



36:52 / 1:12:28

Range >





Question

(iv) A rocket works on the principle of conservation of :

- (a) Linear momentum
- (b) Angular momentum
- (c) Charge
- (d) Force



Rocket Propulsion

- The force which oppose the relative motion



Full screen



Question



[4]

(viii) Two systems are said to be in thermal equilibrium if they have the same:

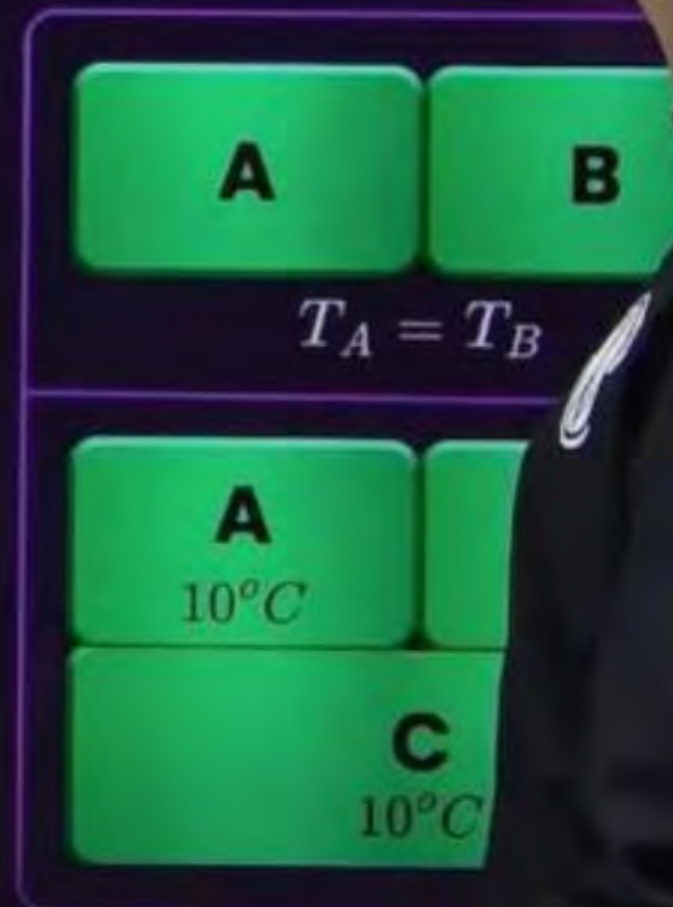
- (a) Pressure
- (b) Volume
- (c) Temperature
- (d) None of them

same temperature

ZEROth LAW OF THERMODYNAMICS

(PYQ)

- ✦ If the temp of A is equal to temp of B and temp of A is equal to temp of C, then temp of B is equal to temp of C automatically.





Question

[5]

(x) The relation between root mean square velocity and absolute temperature is:

(a) $C^2 \propto \sqrt{T}$

(b) $C^2 \propto T^2$

(c) $C \propto \sqrt{T}$

(d) None of them

ROOT MEAN SQUARE VELOCITY



$$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2}{n}}$$

$$v_{rms} = \sqrt{\frac{3RT}{m}}$$

Homework

✓ A Physical quantity is expressed in terms of:-

- 1 Numbers only
- 2 Units only
- 3 Numbers and units
- 4 Symbols only

**Thank
You**